Notes: Nunn (QJE, 2007)

Relationship-Specificity, Incomplete Contracts, and the Pattern of Trade

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1 Big picture

This paper asks whether a country's ability to enforce contracts is a source of comparative advantage. The answer is yes: countries with stronger contract enforcement export relatively more in industries that rely heavily on relationship-specific intermediate inputs.

The economic mechanism is a hold-up problem. When suppliers must make investments that are valuable mainly inside one buyer-seller relationship, weak courts make contracts hard to enforce. Anticipating ex post bargaining or expropriation, suppliers underinvest. Countries with better enforcement suffer less underinvestment, so they have lower production costs in goods that intensively use relationship-specific inputs.

The empirical contribution is to convert this theoretical mechanism into a cross-country, cross-industry test. The paper builds an industry-level measure of contract intensity from input-output tables and Rauch's classification of input market thickness, then asks whether this industry characteristic interacts positively with country-level judicial quality in explaining exports.

Main result. The interaction between contract intensity and judicial quality is positive, statistically significant, and economically large. In the baseline standardized estimates, contract enforcement explains more of the global pattern of trade than physical capital and skilled labor endowments combined.

2 Incremental contribution of the paper

2.1 Relative to traditional trade theory

Traditional comparative advantage emphasizes technology differences, physical capital, skill endowments, land, and other factor endowments. Nunn adds a legal-institutional source of comparative advantage: the capacity to enforce contracts when production requires relationship-specific input investments.

2.2 Relative to the institutions-and-trade literature

Earlier work shows that institutional quality affects trade volumes or broad specialization patterns. This paper narrows the channel to one specific mechanism: incomplete contracts create underinvestment in relationship-specific inputs, and good contract enforcement reduces that distortion.

2.3 Measurement contribution

The paper constructs a new good-level measure of contract intensity:

- use U.S. input-output tables to identify which intermediate inputs enter each final good and with what input shares;
- use Rauch's classification to identify whether inputs are traded on organized exchanges, reference priced, or neither;
- classify inputs not traded on organized exchanges, and in a broader version not reference priced, as more relationship-specific.

This converts an incomplete-contracting idea into a measurable industry variable.

2.4 Empirical-design contribution

The paper uses a Rajan-Zingales style interaction design:

$$\ln x_{ic} = \alpha_i + \alpha_c + \beta_1 z_i Q_c + \beta_2 h_i H_c + \beta_3 k_i K_c + \varepsilon_{ic},$$

where x_{ic} is exports by country c in industry i, z_i is contract intensity, and Q_c is judicial quality. Country fixed effects absorb a country's overall tendency to trade; industry fixed effects absorb global differences across goods. Identification comes from whether high-judicial-quality countries tilt exports toward high-contract-intensity industries.

2.5 Identification contribution

The paper recognizes that judicial quality may be endogenous to trade patterns. It therefore supplements OLS with two strategies:

- instrumental variables using legal origins as instruments for judicial quality;
- propensity-score matching comparing British common-law countries with similar French civil-law countries.

The IV strategy supports the positive effect but raises concerns because over-identification tests reject. The matching exercise is therefore especially important: it provides a more transparent comparison of legally different countries with similar observable characteristics.

3 Core theory and mechanism

3.1 Relationship-specific inputs

An input is relationship-specific when it is worth more inside a particular buyer-seller relationship than outside it. Examples include customized parts, components designed for a particular production process, and inputs that require buyer-specific adaptation.

The relevant incomplete-contracting logic is:

1. A supplier must invest before trade occurs.
2. The investment is difficult to fully describe or enforce in a formal contract.
3. After the investment is sunk, the buyer and seller bargain over surplus.
4. The supplier anticipates ex post hold-up and underinvests ex ante.
5. Better courts reduce the severity of the hold-up problem.

Interpretation: Strong contract enforcement is not just a background condition for all trade. It is especially valuable in goods whose production requires customized intermediate inputs.

3.2 From micro mechanism to comparative advantage

Let z_i denote the importance of relationship-specific inputs in industry i. Let Q_c denote the quality of contract enforcement in country c. The prediction is:

$$\frac{\partial \ln x_{ic}}{\partial (z_i Q_c)} > 0.$$

Countries with high Q_c should specialize in high-z_i goods. Countries with weak Q_c should specialize more in goods using standardized, thick-market inputs.

4 Data and measurement

4.1 Trade and production data

The main dependent variable is total exports by country c in industry i to the rest of the world in 1997. The trade data are from Feenstra's World Trade Flows Database and are concorded into the BEA input-output industry classification. The main sample includes 222 industries and up to 146 countries.

Production data are used only for descriptive raw-data checks because they are available for fewer countries and more aggregated industries.

4.2 Judicial quality and contract enforcement

The baseline country-level measure Q_c is Kaufmann, Kraay, and Mastruzzi's "rule of law" measure, which captures perceptions of the effectiveness and predictability of the judiciary and the enforceability of contracts. The paper also checks robustness using legal quality and World Bank Doing Business measures such as number of procedures, official cost, and time required to collect an overdue debt.

4.3 Factor endowments and factor intensities

To compare the importance of contract enforcement with standard factor-endowment explanations, the paper uses:

- country physical capital per worker, K_c;
- country human capital, H_c;
- industry capital intensity, k_i;
- industry skill intensity, h_i.

These are interacted as $k_i K_c$ and $h_i H_c$ in the trade regressions.

4.4 Contract intensity

The key measurement object is the share of intermediate inputs in industry i that are relationship-specific. Let θ_{ij} be the share of input j in industry i's total intermediate input use. Let R_j^{neither} be

the fraction of input j that is neither traded on an organized exchange nor reference priced. Let R_j^{ref} be the fraction that is reference priced but not exchange traded. The two measures are:

$$z_i^{rs1} = \sum_j \theta_{ij} R_j^{\text{neither}},$$

$$z_i^{rs2} = \sum_j \theta_{ij} \left(R_j^{\text{neither}} + R_j^{\text{ref}} \right).$$

The first measure is stricter: only inputs with neither exchange trading nor reference pricing are classified as relationship-specific. The second measure is broader: reference-priced inputs are also treated as relationship-specific.

Interpretation: The measure uses market thickness as an empirical proxy for relationship specificity. Thick input markets reduce hold-up because a supplier can sell elsewhere; thin markets make investments more relationship-specific.

5 Empirical specification

The main estimating equation is:

$$\ln x_{ic} = \alpha_i + \alpha_c + \beta_1 z_i Q_c + \beta_2 h_i H_c + \beta_3 k_i K_c + \Gamma' W_{ic} + \varepsilon_{ic}.$$

The variables are:

- x_{ic}: exports of country c in industry i;
- z_i: contract intensity of industry i;
- Q_c: judicial quality of country c;
- $h_i H_c$: skill-intensity interacted with skill abundance;
- $k_i K_c$: capital-intensity interacted with capital abundance;
- W_{ic}: additional interaction controls such as income interacted with value added, intra-industry trade, TFP growth, input variety, and credit/GDP interacted with capital intensity.

Interpretation of β_1: A positive β_1 means countries with better contract enforcement export relatively more in industries where relationship-specific inputs are important.

6 Identification and empirical design

6.1 What the fixed effects accomplish

Country fixed effects absorb any country-level forces that affect total exports across all industries, such as size, geography, openness, or overall institutional quality. Industry fixed effects absorb global industry-level forces that affect exports across all countries, such as tradability or global demand.

The identifying variation is therefore within country and within industry: do better-enforcement countries tilt exports toward more contract-intensive industries?

6.2 Key threats

The paper emphasizes two main threats:

- **Omitted comparative-advantage channels.** Contract-intensive industries may also be skill-intensive or technology-intensive, and high judicial-quality countries may also be rich or skill-abundant.
- **Reverse causality.** Countries that specialize in contract-intensive industries may develop better legal institutions because their firms demand them.

6.3 How the paper addresses these threats

The main response to omitted variables is to add standard factor-endowment interactions and many additional industry-country interactions. The main response to reverse causality is to use legal origins as instruments and to compare matched British common-law and French civil-law country pairs.

Interpretation: The empirical design does not claim that judicial quality is randomly assigned. Instead, it builds a set of increasingly demanding tests showing that the contract-enforcement interaction survives many controls and alternative comparisons.

7 Tables and figures

7.1 Table I. Relationship between contract, skill, and capital intensities and Table II. Least and most contract-intensive industries.

TABLE I
RELATIONSHIP BETWEEN CONTRACT, SKILL, AND CAPITAL INTENSITIES

	Contract intensity	
	z_i^{rs1}	z_i^{rs2}
Contract intensity: z_i^{rs2}	.65*	
Skill intensity: h_i	.44*	.28*
Capital intensity: k_i	−.49*	−.38*

Correlation coefficients are reported. * indicates significance at the 1 percent level.

Table I. Relationship between contract, skill, and capital intensities.

TABLE II
THE TWENTY LEAST AND TWENTY MOST CONTRACT INTENSIVE INDUSTRIES

Least contract intensive: lowest z_i^{rs1}		Most contract intensive: highest z_i^{rs1}	
z_i^{rs1}	Industry description	z_i^{rs1}	Industry description
.024	Poultry processing	.810	Photographic & photocopying equip. manuf.
.024	Flour milling	.819	Air & gas compressor manuf.
.036	Petroleum refineries	.822	Analytical laboratory instr. manuf.
.036	Wet corn milling	.824	Other engine equipment manuf.
.053	Aluminum sheet, plate & foil manuf.	.826	Other electronic component manuf.
.058	Primary aluminum production	.831	Packaging machinery manuf.
.087	Nitrogenous fertilizer manufacturing	.840	Book publishers
.099	Rice milling	.851	Breweries
.111	Prim. nonferrous metal, excl. copper & alum.	.854	Musical instrument manufacturing
.132	Tobacco stemming & redrying	.872	Aircraft engine & engine parts manuf.
.144	Other oilseed processing	.873	Electricity & signal testing instr. manuf.
.171	Oil gas extraction	.880	Telephone apparatus manufacturing
.173	Coffee & tea manufacturing	.888	Search, detection, & navig. instr. manuf.
.180	Fiber, yarn, & thread mills	.891	Broadcast & wireless comm. equip. manuf.
.184	Synthetic dye & pigment manufacturing	.893	Aircraft manufacturing
.190	Synthetic rubber manufacturing	.901	Other computer peripheral equip. manuf.
.195	Plastics material & resin manuf.	.904	Audio & video equipment manuf.
.196	Phosphatic fertilizer manufacturing	.956	Electronic computer manufacturing
.200	Ferroalloy & related products manuf.	.977	Heavy duty truck manufacturing
.200	Frozen food manufacturing	.980	Automobile & light truck manuf.

The contract intensity measures reported are rounded from seven digits to three digits.

Table II. Least and most contract-intensive industries.

Read:

- Table I shows that the two contract-intensity measures are strongly correlated with each other: 0.65.
- Contract intensity is positively correlated with skill intensity (0.44 for z_i^{rs1} and 0.28 for z_i^{rs2}), but negatively correlated with capital intensity (−0.49 and −0.38).
- Table II shows that low-contract-intensity industries are standardized input industries such as poultry processing, flour milling, petroleum refining, and aluminum production.
- High-contract-intensity industries include aircraft, computers, electronic components, engines, compressors, and automobiles.

Economic message: The measurement has face validity. Industries that need customized, com-

plex, or highly engineered inputs appear at the high end of the contract-intensity distribution, while standardized commodity-like industries appear at the low end. Table I also warns that contract intensity is correlated with skill intensity, so controlling for human capital is essential.

7.2 Table III. Judicial quality and average contract intensity of production and exports and Table IV. Main determinants of comparative advantage.

TABLE III
JUDICIAL QUALITY AND THE AVERAGE CONTRACT INTENSITY OF PRODUCTION AND OF EXPORTS

	Output regressions		Export regressions	
	$\bar{Z}_c^{rs1}$	$\bar{Z}_c^{rs2}$	$\bar{Z}_c^{rs1}$	$\bar{Z}_c^{rs2}$
Judicial quality: Q_c	.392**	.465**	.290**	.291**
	(.109)	(.109)	(.081)	(.065)
Number of obs.	78	78	146	146
R^2	.15	.22	.08	.08

The dependent variables are the average contract intensity of production or exports. Standardized beta coefficients are reported, with robust standard errors in brackets. ** indicates significance at the 1 percent level.

Table III. Judicial quality and average contract intensity of production and exports.

TABLE IV
THE DETERMINANTS OF COMPARATIVE ADVANTAGE

	(1)	(2)	(3)	(4)	(5)
Judicial quality interaction: z_iQ_c	.289**	.318**	.326**	.235**	.296*
	(.013)	(.020)	(.023)	(.017)	(.024
Skill interaction: h_iH_c			.085**		.063*
			(.017)		(.017
Capital interaction: k_iK_c			.105**		.074
			(.031)		(.041
Log income × value added: $va_i \ln y_c$				−.117*	−.137
				(.047)	(.067
Log income × intra-industry trade: $iit_i \ln y_c$				.576**	.546*
				(.041)	(.056
Log income × TFP growth: $\Delta tfp_i \ln y_c$				.024	−.01
				(.033)	(.049
Log credit/GDP × capital: k_iCR_c				.020	.021
				(.012)	(.018
Log income × input variety: $(1 - hi_i) \ln y_c$				.446**	.522*
				(.075)	(.103
Country fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	.72	.76	.76	.77	.76
Number of observations	22,598	10,976	10,976	15,737	10,81

Dependent variable is $\ln x_{ic}$. The regressions are estimates of (1). The dependent variable is the natural log of exports in industry i by country c to all other countries. In a
regressions the measure of contract intensity used is z_i^{rs1}. Standardized beta coefficients are reported, with robust standard errors in brackets. * and ** indicate significance at th
5 and 1 percent levels.

Table IV. Main determinants of comparative advantage.

Read:

- Table III shows positive raw correlations between judicial quality and the average contract intensity of both output and exports.
- The coefficient on judicial quality is 0.392 for output using z_i^{rs1} and 0.290 for exports using z_i^{rs1}; all estimates in the table are significant at the 1 percent level.
- Table IV reports the main export regressions. The judicial quality interaction z_iQ_c is positive and significant in every specification.
- With factor endowment controls, the standardized coefficient on z_iQ_c is 0.326, compared with 0.085 for skill and 0.105 for capital.
- With the full set of controls, the coefficient on z_iQ_c remains 0.296.

Economic message: The main results support the paper's core claim: better contracting institutions create comparative advantage in contract-intensive goods. The standardized coefficients imply that the contract channel is not a small residual institutional effect; in the main specifications it is larger than the combined standard factor-endowment channels.

7.3 Table V. Additional results and Table VI. Robustness and sensitivity analysis

TABLE V
ADDITIONAL RESULTS

	(1)	(2)	(3)	(4)	(5)
Judicial quality interaction: $z_i Q_c$	.186**	.324**	.253**	.177**	.236*
	(.015)	(.026)	(.030)	(.025)	(.021
Industries with many inputs: $z_i Q_c I_i^{n_i > \bar{n}}$	.156**				
	(.011)				
Skill endowment × contract intensity: $z_i H_c$		.049*			
		(.020)			
Capital endowment × contract intensity: $z_i K_c$			.181**		
			(.036)		
Log income × contract intensity: $z_i \ln y_c$				.310**	
				(.060)	
Log credit/GDP × contract intensity: $z_i CR_c$					.077*
					(.016
Country fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	.73	.72	.72	.72	.73
Number of observations	22,598	13,201	13,201	21,769	19,24

The regressions are estimates of (1). The dependent variable is the natural log of exports in industry i by country c to all other countries, $\ln x_{ic}$. In all regressions the measu of contract intensity used is z_i^{rs1}. Standardized beta coefficients are reported, with robust standard errors in brackets. * and ** indicate significance at the 5 and 1 percent level

Table V. Additional results.

TABLE VI
ROBUSTNESS AND SENSITIVITY ANALYSIS

	Without factor endowment controls		With factor endowment controls	
	z_i^{rs1}	z_i^{rs2}	z_i^{rs1}	z_i^{rs2}
Using alternative measures of Q_c:				
Legal quality	.360**	.446**	.370**	.358**
	(.016)	(.025)	(.025)	(.037)
	19,072	19,072	10,824	10,824
Number of procedures	.179**	.231**	.244**	.261**
	(.015)	(.025)	(.024)	(.039)
	18,873	18,873	10,288	10,288
Official costs	.294**	.466**	.250**	.353**
	(.020)	(.031)	(.031)	(.046)
	18,873	18,873	10,288	10,288
Time	.191**	.213**	.160**	.161**
	(.021)	(.030)	(.027)	(.040)
	18,873	18,873	10,288	10,288
Using "rule of law" to measure Q_c:				
Outliers omitted	.264**	.432**	.327**	.428**
	(.011)	(.018)	(.019)	(.028)
	21,409	21,389	10,378	10,379
OECD countries	.481**	.343**	.518**	.223*
	(.069)	(.087)	(.081)	(.103)
	4,941	4,941	3,906	3,906
Non-OECD countries	.232**	.389**	.251**	.348**
	(.018)	(.029)	(.032)	(.049)
	17,657	17,657	7,070	7,070
Using 1963 data	.441**	.536**	.288**	.290**
	(.042)	(.069)	(.049)	(.086)
	5,997	5,997	5,322	5,322

The regressions are estimates of (1). The dependent variable is the natural log of exports in industry i by country c to all other countries, $\ln x_{ic}$. Each entry of the table reports the estimated beta coefficients for $z_i Q_c$, with robust standard errors reported in brackets. Below this the number of observations in the regression is reported. * and ** indicate significance at the 5 and 1 percent levels.

Table VI. Robustness and sensitivity analysis.

Read:

- Table V adds tests for vertical integration and alternative country characteristics.
- The baseline judicial-quality interaction remains positive in all columns.
- In column (1), the interaction with industries using many inputs is positive (0.156), meaning the judicial-quality effect is stronger where vertical integration is less feasible.
- Table VI checks alternative contract-enforcement measures, alternative contract-intensity definitions, outlier exclusion, OECD and non-OECD samples, and 1963 data.
- Across these exercises, the estimated judicial-quality interaction remains positive and statistically significant.

Economic message: Table V strengthens the mechanism: when many inputs are used, internalizing all relationships through vertical integration is harder, so courts matter more. Table VI shows that the result is not an artifact of one particular contract-enforcement measure, one sample, or one time period.

7.4 Table VII. IV estimates using legal origins as instruments and Table VIII. Matched British common-law and French civil-law countries.

TABLE VII
IV Estimates using Legal Origins as Instruments

	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)
OLS and second stage IV estimates: Dependent variable is ln x_{ic}.						
Judicial quality interaction: z_iQ_c	.288**	.385**	.326**	.539**	.296**	.520**
	(.013)	(.022)	(.023)	(.044)	(.024)	(.046)
Skill interaction: h_iH_c			.085**	.042*	.063**	.023
			(.017)	(.019)	(.017)	(.019)
Capital interaction: h_iK_c			.105**	.183**	.074	.114**
			(.031)	(.035)	(.041)	(.043)
Full set of control variables	No	No	No	No	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	.72	.72	.76	.76	.76	.76
Number of observations	22,598	22,598	10,976	10,976	10,816	10,816
First stage IV estimates: Dependent variable is z_iQ_c.						
British legal origin: z_iB_c		−.295**		−.210**		−.215**
		(.033)		(.038)		(.036)
French legal origin: z_iF_c		−.405**		−.304**		−.298**
		(.025)		(.030)		(.028)
German legal origin: z_iG_c		−.072		−.072		−.088
		(.045)		(.051)		(.049)
Socialist legal origin: z_iS_c		−.477**				
		(.035)				
F-test		113.1		74.6		60.4
Hausman test (p-value)		.00		.00		.00
Over-id test (p-value)		.00		.00		.00

In the second stage standardized beta coefficients are reported, with robust standard errors in brackets. The dependent variable is the natural log of exports in industry i by country c to all other countries. In the first stage I report regular coefficients, with robust standard errors clustered at the country level reported in brackets. The dependent variable is the judicial quality interaction z_iQ_c. The measure of contract intensity used is z_i^{74}. Although all explanatory variables in the second stage are also included in the first stage, to conserve space I do not report the first stage coefficients for these variables. The omitted legal origin category is Scandinavian. Because there are no Socialist legal origin countries in the smaller samples of columns (3)-(5), the Socialist interaction term does not appear as an instrument in these specifications. The reported F-test is for the null hypothesis that the coefficients for the interaction terms are jointly equal to zero. * and ** indicate significance at the 5 and 1 percent levels.

Table VII. IV estimates using legal origins as instruments.

TABLE VIII
Comparing Matched British Common Law and French Civil Law Countries

	(1)	(2)	(3)	(4)	(5)	(6)
		Matched by				
	Not matched	Per-cap. GDP	Financial develop.	Factor endow.	Trade open.	All vars.
Contract intensity: z_i	1.17**	.491**	.491**	2.13**	1.22**	.855**
	(.156)	(.191)	(.188)	(.248)	(.238)	(.216)
Country pair fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	.70	.57	.68	.55	.69	.67
Number of obs.	386,872	5,234	4,995	4,329	3,627	3,982

The regressions are estimates of (2). The dependent variable is the natural log of the ratio of the common law country's total exports in industry i relative to the civil law country's total exports in industry i, $\ln(x_{ij}/x_{i'j})$. The measure of contract intensity used is z_i^{74}. Estimated coefficients are reported, with robust standard errors adjusted for clustering at the industry level in brackets. * and ** indicate significance at the 5 and 1 percent levels.

Table VIII. Matched British common-law and French civil-law countries.

Read:

- Table VII instruments judicial quality using legal-origin interactions. The IV estimates for z_iQ_c are positive and significant: 0.385, 0.539, and 0.520 across the reported IV columns.
- The first-stage relationship is strong, with large F-statistics.
- However, the over-identification tests reject, suggesting that legal origin may affect comparative advantage through channels other than judicial quality.
- Table VIII compares British common-law countries with French civil-law countries, including matched pairs based on income, financial development, factor endowments, trade openness, and all variables.
- The contract-intensity coefficient remains positive and significant across all matching columns; with all variables matched, it is 0.855.

Economic message: The IV evidence supports the direction and significance of the contract channel, but the paper treats the exclusion restriction with caution. The matching evidence is useful because it compares more similar countries and still finds that common-law countries export more in contract-intensive industries than comparable French civil-law countries.

8 The main coefficient

The key coefficient is the interaction z_iQ_c. It is not saying simply that countries with better courts export more, because country fixed effects absorb that. It is not saying simply that contract-intensive goods are traded more, because industry fixed effects absorb that. It says that better-court countries export relatively more in the industries where contract enforcement should matter most.

For example, moving from a weak-contract-enforcement country to a strong-contract-enforcement country should have a larger predicted effect on exports of electronic computers, aircraft, or automobiles than on exports of flour, poultry, or petroleum refining.

9 Why the result matters

The paper changes the interpretation of institutions in trade. Institutions are not merely broad background features that raise all activity. They shape the pattern of specialization. A country with strong legal institutions has a comparative advantage in goods that require complex contractual relationships among input suppliers and final producers.

This mechanism also helps explain why developing countries may struggle to enter technologically sophisticated industries even when they have cheap labor. If those industries rely on relationship-specific inputs, weak contract enforcement raises costs by discouraging supplier investment.

10 Limitations and caveats

- **U.S. input-output structure.** The paper uses U.S. input-output tables to measure input shares for all countries. This is necessary for comparability, but it assumes that global production technologies resemble the U.S. structure.
- **Measurement of relationship specificity.** Rauch's exchange/reference-price classification is an indirect proxy for relationship-specific investment. Thin markets are plausibly related to specificity, but the measure is not perfect.
- **Legal-origin exclusion restriction.** Legal origins may affect financial development, investor protection, or other institutions, not only contract enforcement. The paper recognizes this and therefore treats IV results cautiously.
- **Positive export observations.** The main regressions condition on positive exports, so they primarily explain export volumes among active exporter-industry pairs rather than entry into exporting.
- **Cross-sectional design.** The paper is mainly cross-sectional around 1997. This design is powerful for patterns of specialization but weaker for tracing dynamic institutional change.

11 Short summary

Nunn (2007) argues that contract enforcement is a source of comparative advantage. The key idea is that incomplete contracts cause underinvestment when inputs require relationship-specific investments. Countries with better legal systems reduce this underinvestment and therefore specialize in contract-intensive goods.

The empirical strategy constructs contract intensity from U.S. input-output tables and Rauch's market-thickness classification, then estimates whether the interaction between contract intensity and judicial quality predicts exports. The answer is robustly yes. The main effect is large relative to standard factor-endowment channels. Legal-origin IV estimates and matched common-law versus civil-law comparisons support the same qualitative conclusion, although the paper is cautious about the IV exclusion restriction.

12 Conclusion

The paper's central lesson is that legal institutions can determine what countries produce and export. Good courts are especially valuable when production requires customized inputs and relationship-specific investments. Thus, the quality of contract enforcement is not only a determinant of the overall level of development; it is also a determinant of the international pattern of specialization.